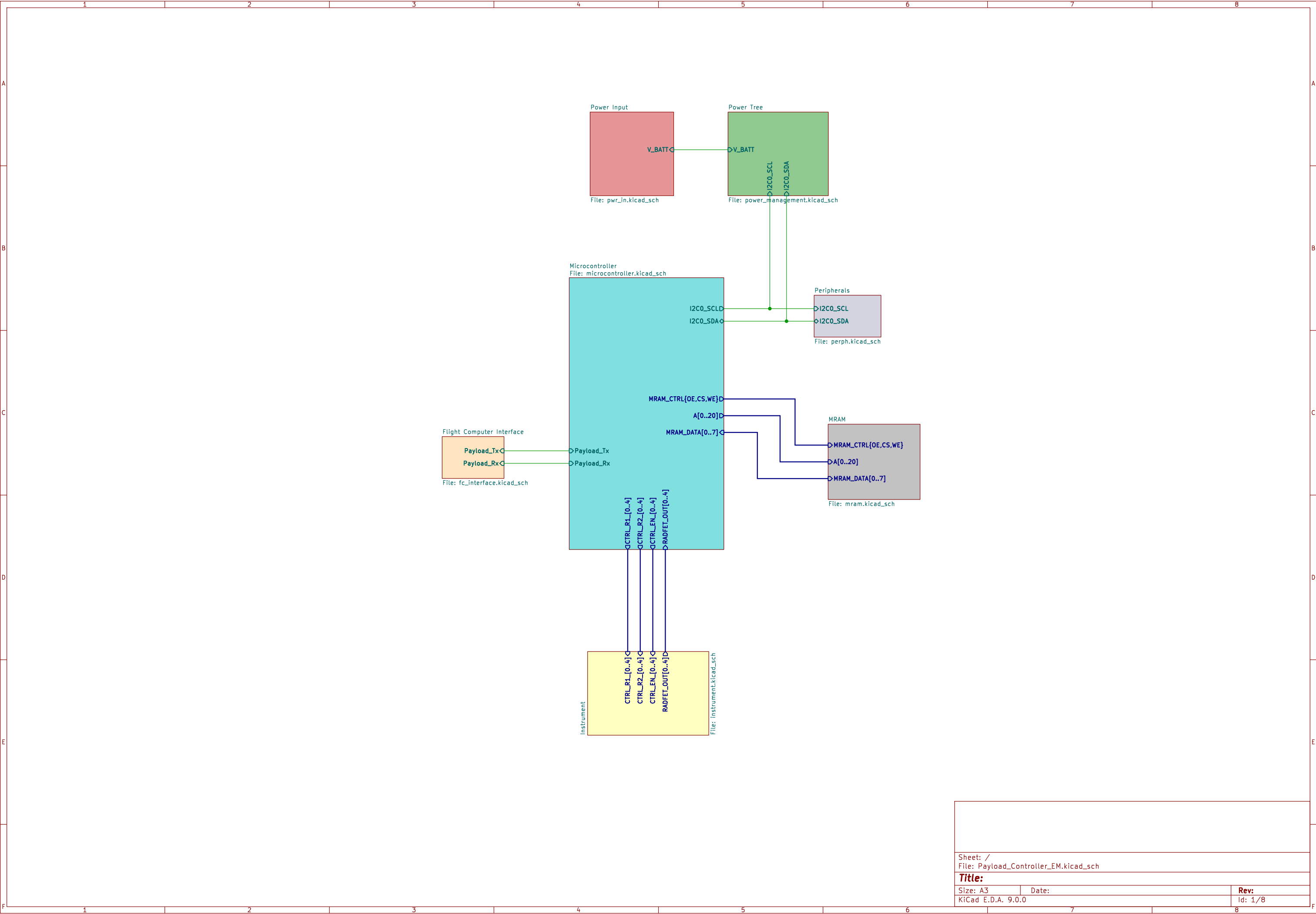
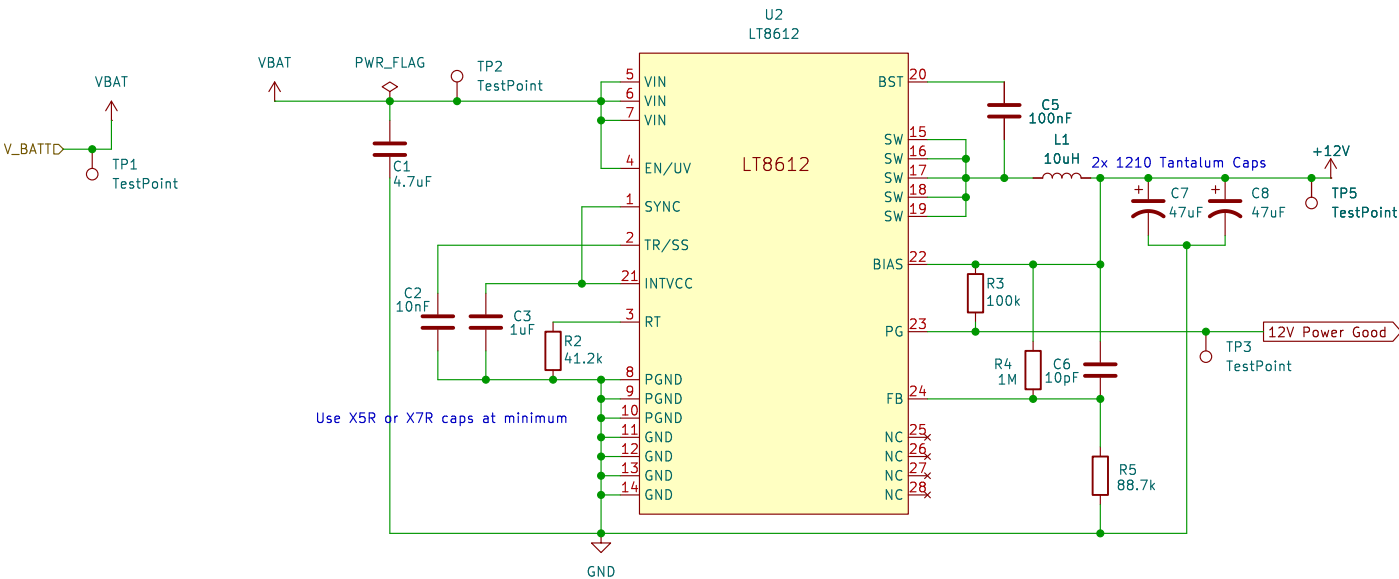
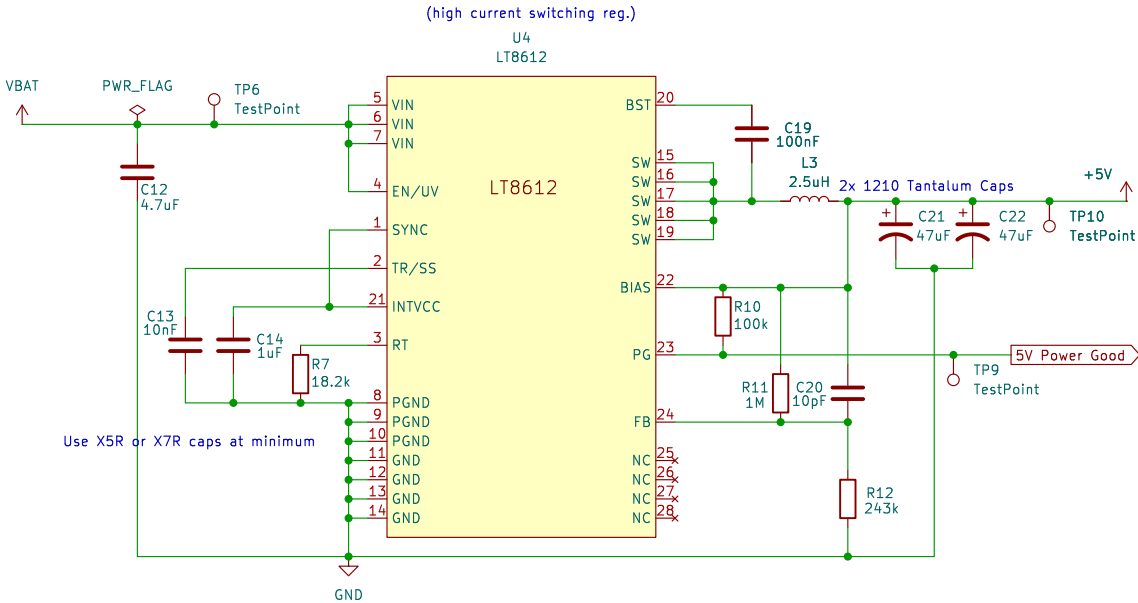




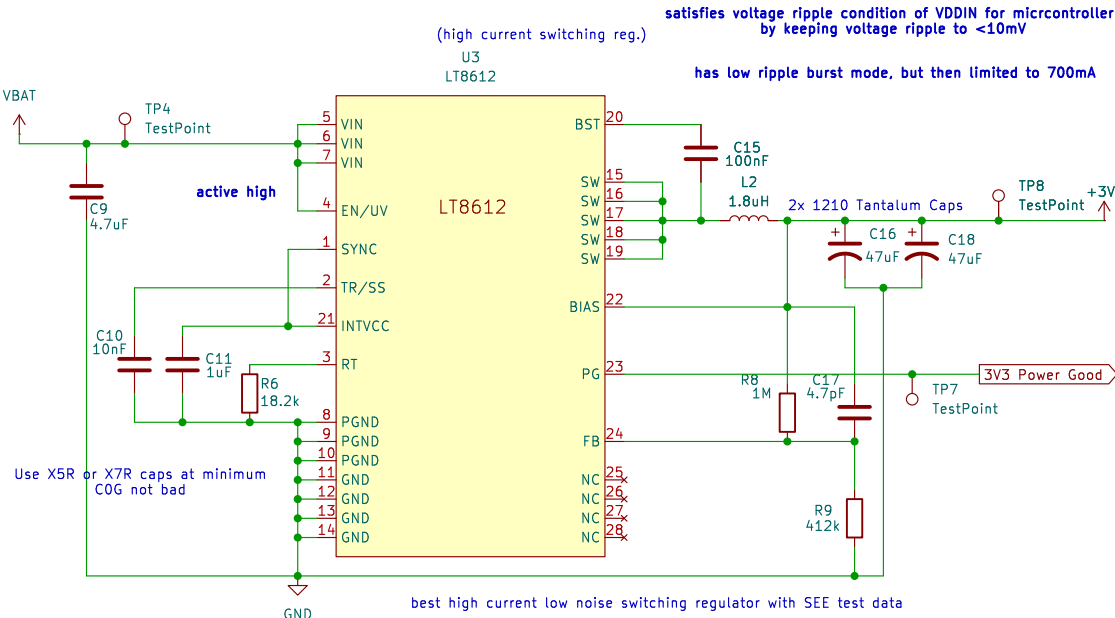
Switching Reg – Vbat to +12V



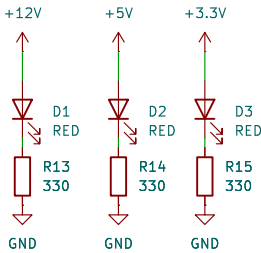
Switching Reg – Vbat to +5V



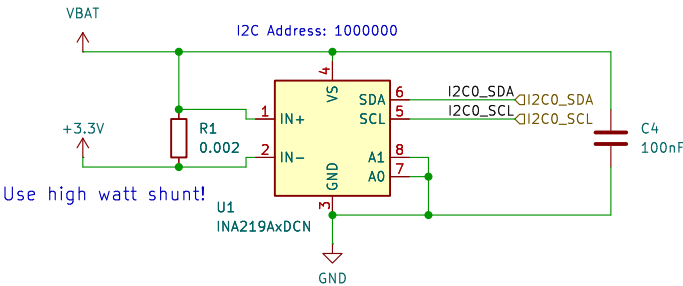
Switching Reg – Vbat to 3.3V



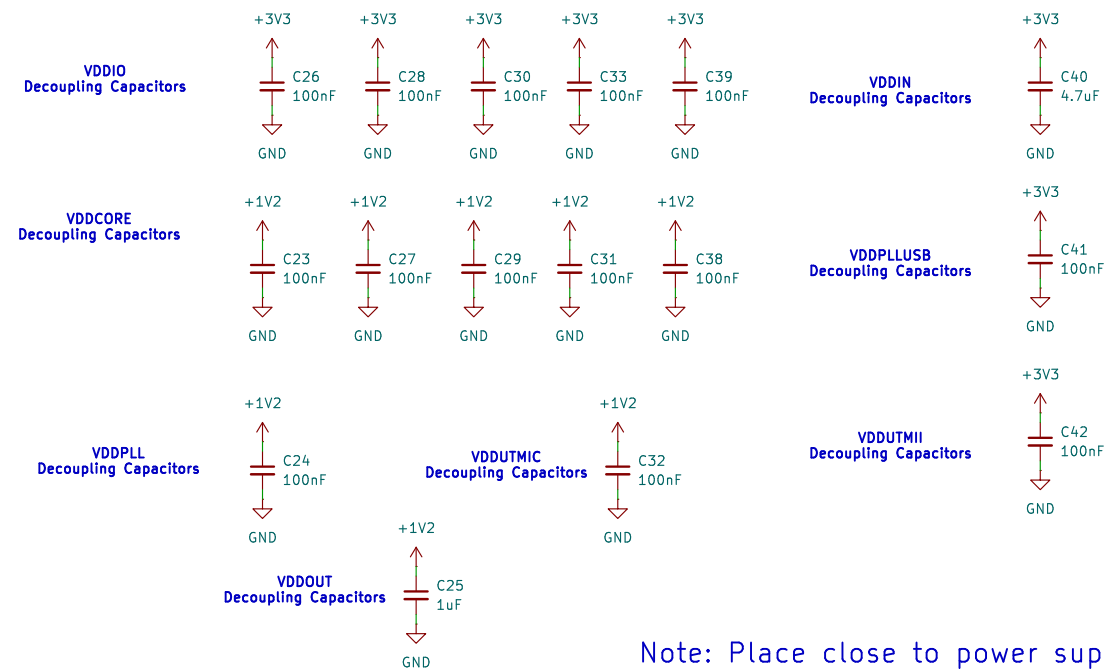
Debug LEDs



Board Power Monitor



Decoupling Capacitors

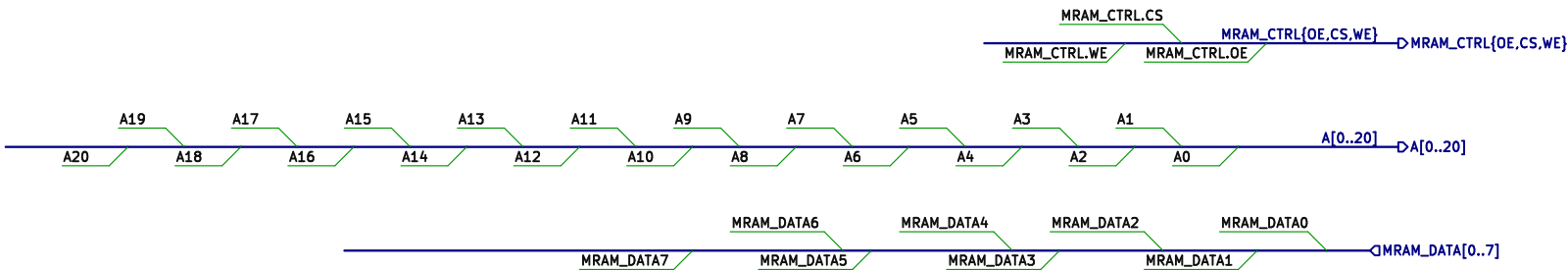


Note: Place close to power supply pins

Table 7-1. Power Supplies		
Name	Associated Ground	Powers
VDDCORE	GND	Core, embedded memories and peripherals.
VDDIO	GND	Peripheral I/O lines (Input/Output Buffers), backup part, 1 Kbytes of backup SRAM, 32 kHz crystal oscillator, oscillator pads. For USB operations, VDDIO voltage range must be between 3.0V and 3.6V.
VDDIN	GND, GNDANA	Voltage regulator input. Supplies also the ADC, DAC, and analog voltage comparator.
VDDPLL	GND, GNDPLL	PLL and the fast RC oscillator.
VDDPLLUSB	GNDPLLUSB	UTMI PLL and 3 MHz to 20 MHz oscillator.
VDDUTMI	GNDUTMI	USB transceiver interface. Must be connected to VDDIO.
VDDUTMIC	GNDUTMI	USB transceiver core.

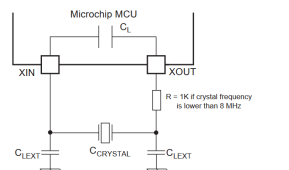
Table 58-5. Voltage Regulator Characteristics						
Symbol	Parameter	Conditions	Min.	Typ.	Max. Unit	
V _{DDOUT}	DC Output Voltage	Normal mode, I _{L,LOAD} = 100 mA	1.2	1.23	1.26	V
		Standby mode	—	0	—	
I _{L,LOAD}	Maximum DC Output Current		—	—	150	mA
C _{DI}	Input Decoupling Capacitor	(1)	—	4.7	—	μF
C _{DO}	Output Decoupling Capacitor	(2)	—	1	—	μF
		ESR	—	2	—	Ohm
t _{ON}	Turn-on Time	C _{DO} = 1 μF, V _{DDOUT} reaches DC output voltage	—	1	2.5	ms

MRAM Bus Lines



Crystal Oscillator

Figure 57-10. 3 to 20 MHz Crystal Oscillator Schematics



where, C_{EXT} is the capacitance of the printed circuit board (PCB) track layout from the crystal to the pin.

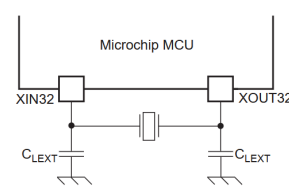
57.4.9. Crystal Oscillator Design Considerations

57.4.9.1. Choosing a Crystal

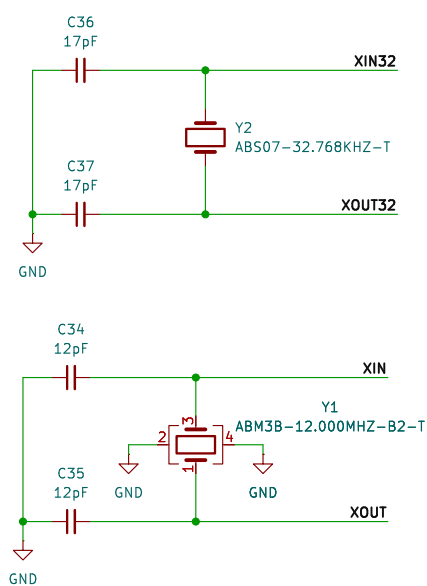
When choosing a crystal for the 32768 Hz Slow Clock Oscillator or for the 3 MHz-20 MHz oscillator, users need to consider several parameters. Important parameters between crystal and product specifications are as follows:

- Crystal Load Capacitance
 - The total capacitance loading the crystal, including the oscillator's internal parasitics and the PCB parasitics, must match the load capacitance for which the crystal's frequency is specified. Any mismatch in the load capacitance with respect to the crystal's specification will lead to inaccurate oscillation frequency.

Figure 57-9. 32.768 kHz Crystal Oscillator Schematics



C_{EXT} = 2 × (C_{CRYSTAL} - C_{PARA} - C_{PCB})



Debug Section

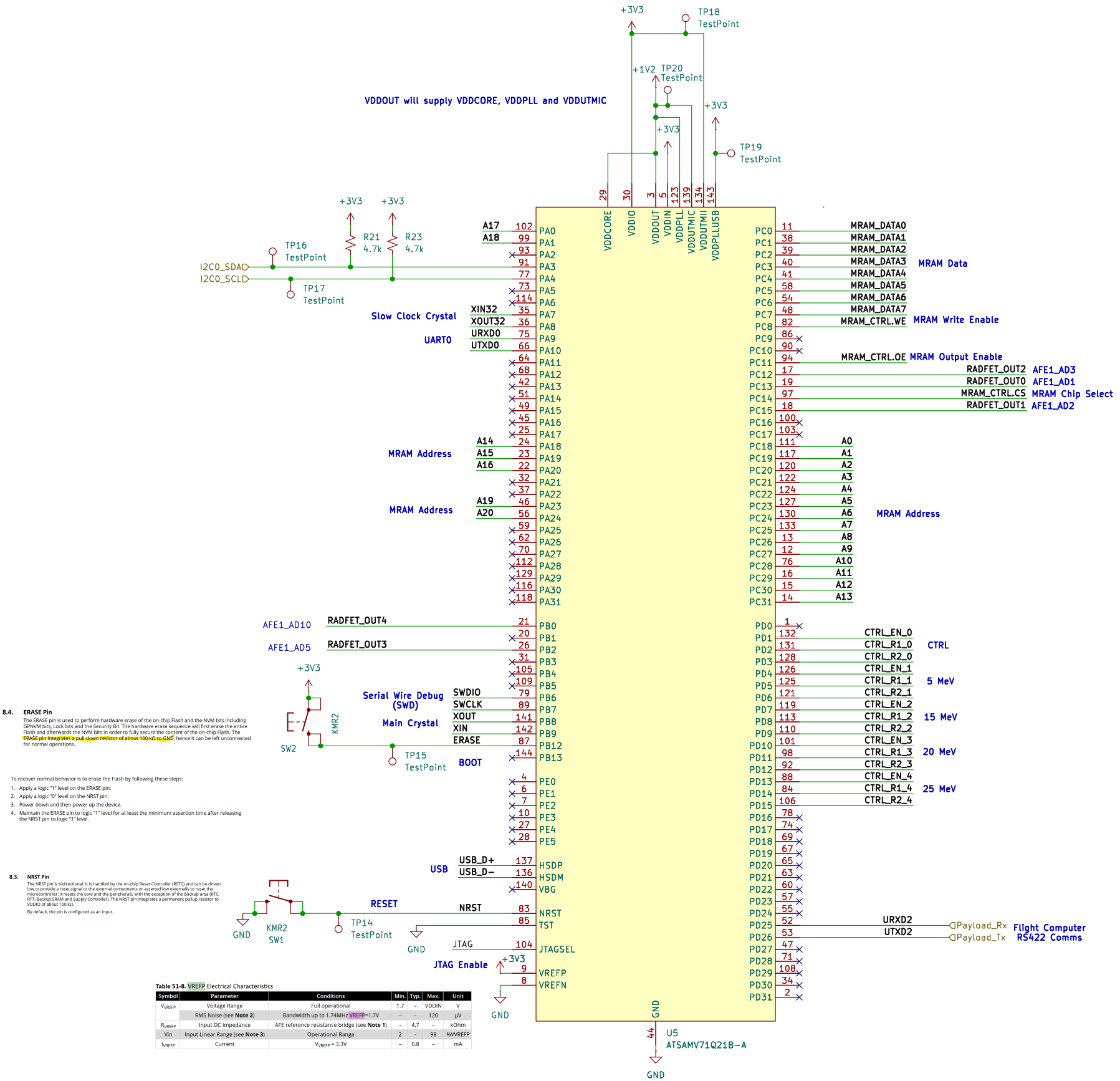
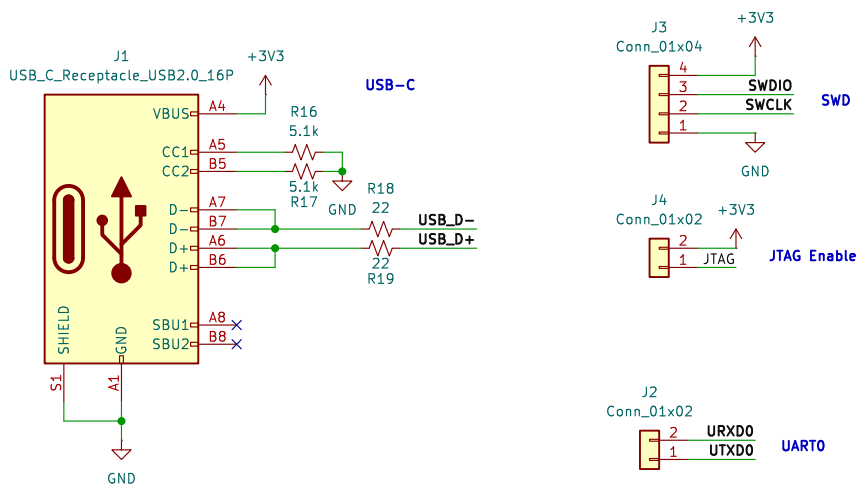
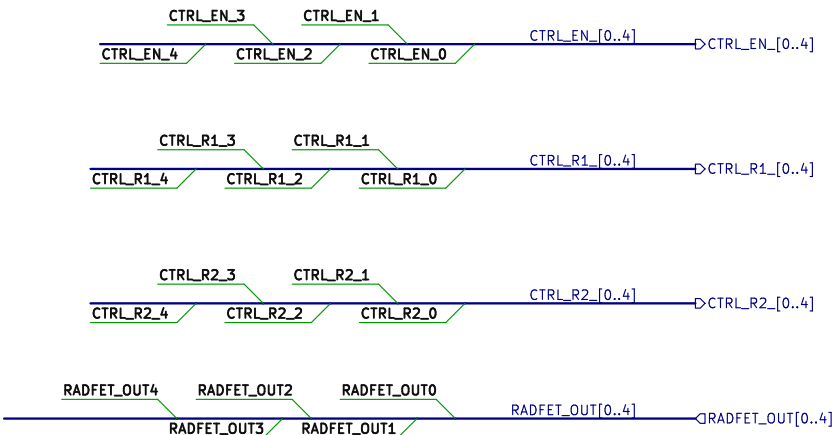
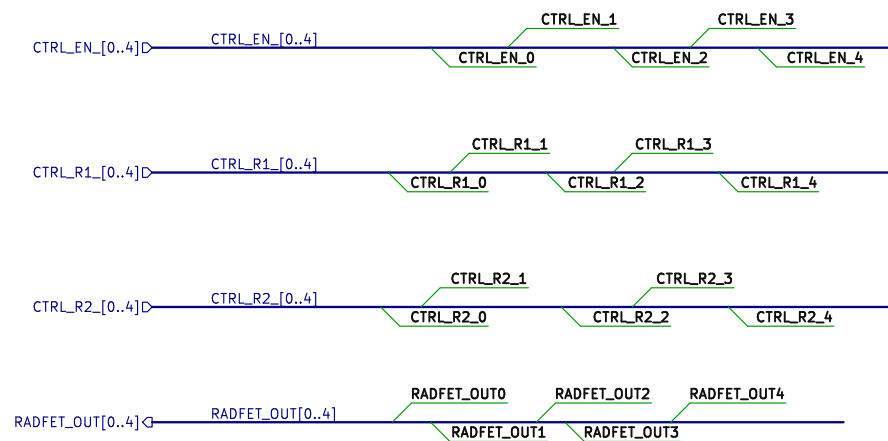


Table 51-4. VREFP Electrical Characteristics					
Symbol	Parameter	Conditions	Min.	Typ.	Max. Unit
VREFP	Voltage Range	Full operational	1.2	-	VDDIO V
	RMS Noise (see Note 2)	Bandwidth up to 5.74MHz; VREFP = 1.7V	-	-	120 μ V
RREFP	Input DC Impedance	AFE reference resistance bridge (see Note 1)	-	4.7	- kOhm
IREFP	Input Linear Range (see Note 3)	Operational Range	2	-	98 μ V/VREFP
IREFP	Current	VREFP = 3.3V	-	0.8	- mA

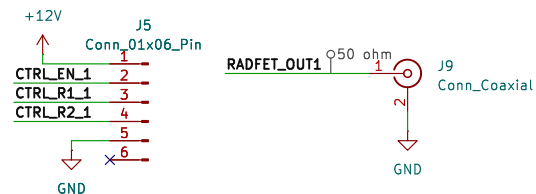
RadFET Bus Lines



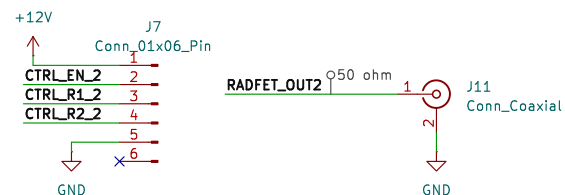
MCU Control Signals



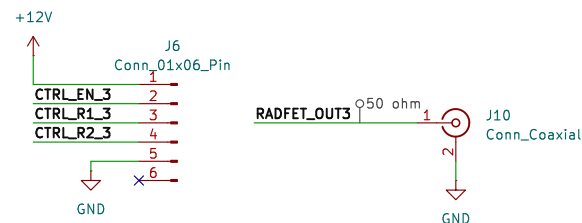
5 MeV



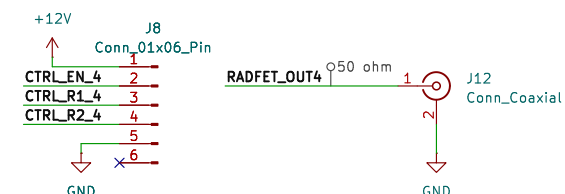
15 MeV



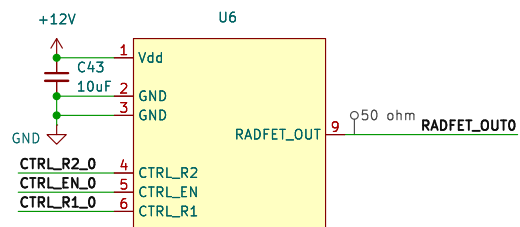
20 MeV



25 MeV



Control Readout Module



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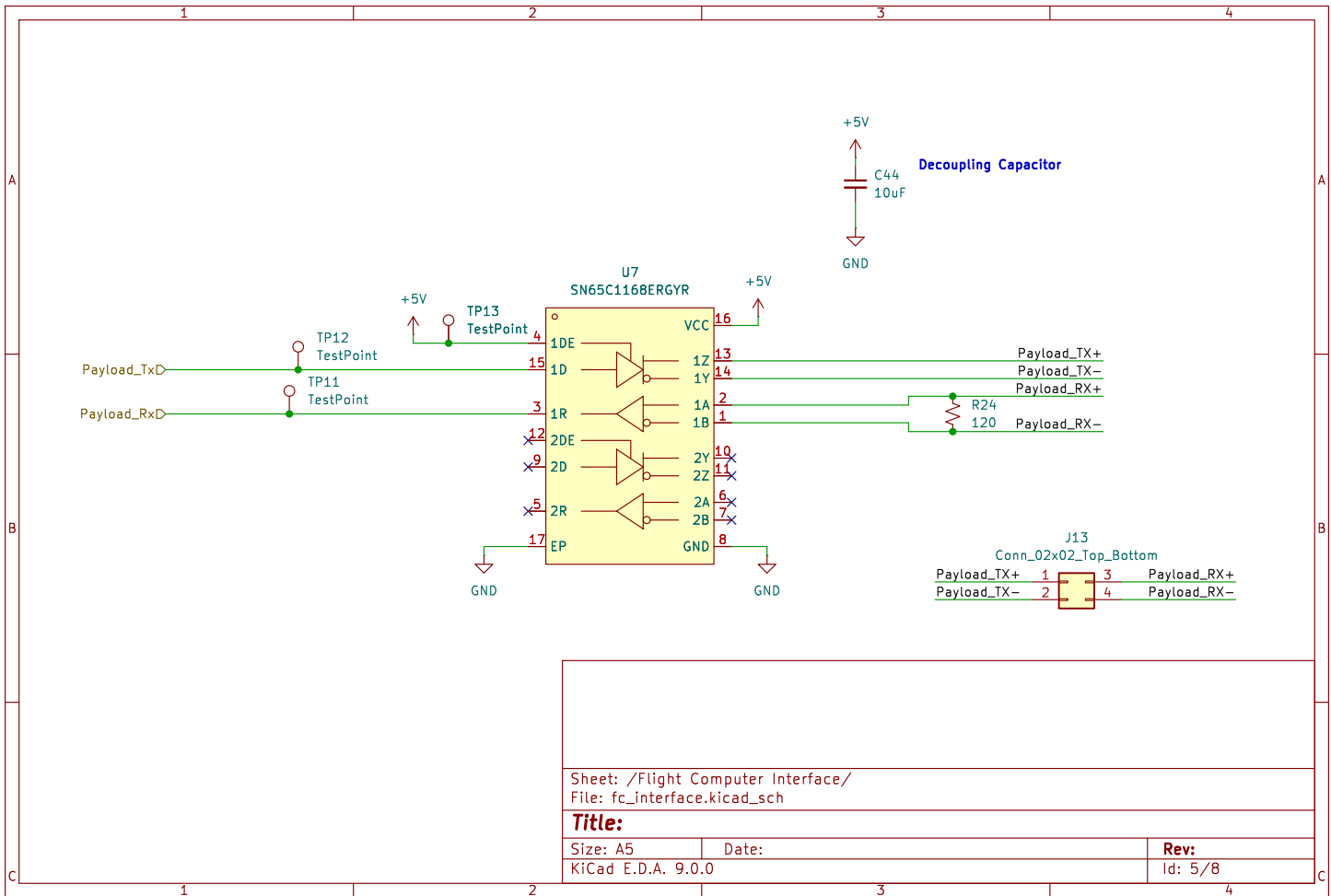
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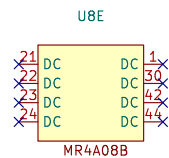
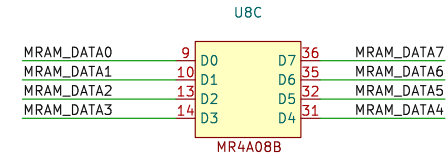
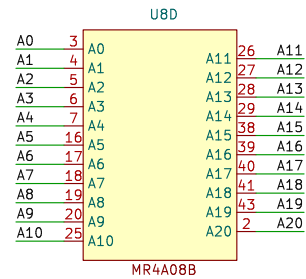
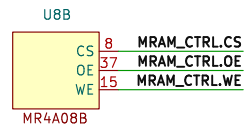
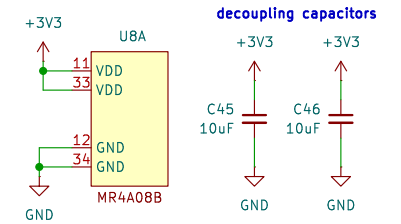
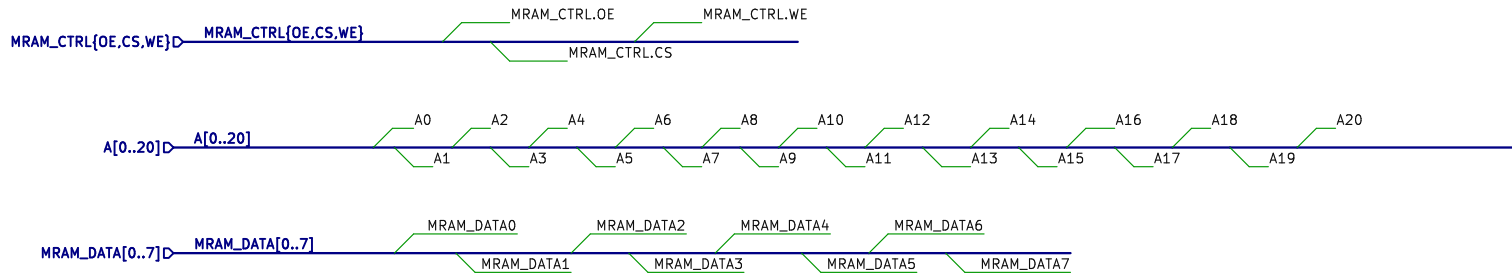
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KiCad E.D.A. 9.0.0

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